

Worksheet: Convolution and Pooling Operations (Chapter 5)

Problem 1: Convolution and Max Pooling Pipeline

Problem Setup: You are given an input tensor $\mathbf{X} \in \mathbb{R}^{4 \times 4 \times 1}$ ($C_{\text{in}} = 1$) and a filter kernel $\mathbf{K} \in \mathbb{R}^{2 \times 2 \times 1}$ ($C_{\text{out}} = 1$), with **Scalar Bias** $b = 0$ and **Activation Function** $\text{ReLU}(z) = \max(0, z)$:

$$\mathbf{X}_{::,1} = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 \end{bmatrix}, \quad \mathbf{K}_{::,1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Part A: Convolution Operation

Assuming stride $s = 1$ and padding $p = 0$, first compute the spatial and channel dimensions of the output feature map, then calculate all values of the output feature map $\mathbf{Y}$ after adding bias $b = 0$ and applying activation $\text{ReLU}(z) = \max(0, z)$.

1. **Output Dimensions** ($H \times W \times C$): _____

2. **Feature Map Calculations & Resulting Matrix:**

Part B: Max Pooling

Apply a 2×2 Max Pooling filter with stride $s = 1$ and padding $p = 0$ to the feature map $\mathbf{Y}$ you computed in Part A.

1. **Output Dimensions** ($H \times W \times C$): _____

2. **Feature Map Calculations & Resulting Matrix:**

Problem 2: Multi-Channel Convolution and Average Pooling Pipeline

Problem Setup: You are given an input tensor $\mathbf{X} \in \mathbb{R}^{3 \times 3 \times 2}$ ($C_{\text{in}} = 2$) and a filter kernel $\mathbf{K} \in \mathbb{R}^{2 \times 2 \times 2}$ ($C_{\text{out}} = 1$) with **Scalar Bias** $b = -1$ and **Activation Function** $\text{ReLU}(z) = \max(0, z)$:

$$\mathbf{X}_{:, :, 1} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{X}_{:, :, 2} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{K}_{:, :, 1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{K}_{:, :, 2} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Part A: Multi-Channel Convolution Operation

Assuming stride $s = 1$ and padding $p = 0$, first compute the spatial and channel dimensions of the output feature map, then calculate all values of the output feature map $\mathbf{Y}$ after adding bias $b = -1$ and applying activation $\text{ReLU}(z) = \max(0, z)$.

- 1. **Output Dimensions** ($H \times W \times C$): _____
- 2. **Feature Map Calculations & Resulting Matrix:**

Part B: Average Pooling

Apply a 2×2 Average Pooling filter with stride $s = 1$ and padding $p = 0$ to the feature map $\mathbf{Y}$ you computed in Part A.

- 1. **Output Dimensions** ($H \times W \times C$): _____
- 2. **Feature Map Calculations & Resulting Matrix:**

Problem 3: Multi-Filter Convolution and Max Pooling with Rectangular Tensors

Problem Setup: You are given a rectangular input tensor $\mathbf{X} \in \mathbb{R}^{3 \times 4 \times 2}$ ($H_{\text{in}} = 3, W_{\text{in}} = 4, C_{\text{in}} = 2$) and **two independent filters** $\mathbf{K}^{(1)}, \mathbf{K}^{(2)} \in \mathbb{R}^{2 \times 2 \times 2}$ ($C_{\text{out}} = 2$) with **Biases** $b_1 = 0, b_2 = -1$ and **Activation Function** $\text{ReLU}(z) = \max(0, z)$:

$$\mathbf{X}_{::,1} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 \end{bmatrix}, \quad \mathbf{X}_{::,2} = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$
$$\mathbf{K}_{::,1}^{(1)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{K}_{::,2}^{(1)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{K}_{::,1}^{(2)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{K}_{::,2}^{(2)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Part A: Multi-Filter Convolution Operation

Assuming stride $s = 1$ and padding $p = 0$, first compute the spatial and channel dimensions of the output feature tensor, then calculate all values of the output feature tensor $\mathbf{Y}$ after adding per-filter biases and applying activation $\text{ReLU}(z) = \max(0, z)$.

- 1. **Output Dimensions** ($H \times W \times C$): _____
- 2. **Feature Map Calculations & Resulting Matrix:**

Part B: Max Pooling

Apply a 2×2 Max Pooling filter with stride $s = 1$ and padding $p = 0$ to each channel of the feature map $\mathbf{Y}$ you computed in Part A.

- 1. **Output Dimensions** ($H \times W \times C$): _____
- 2. **Feature Map Calculations & Resulting Matrix:**

Problem 4: Convolution and Max Pooling with Stride $s = 2$

Problem Setup: You are given an input tensor $\mathbf{X} \in \mathbb{R}^{5 \times 5 \times 1}$ ($C_{\text{in}} = 1$) and a filter kernel $\mathbf{K} \in \mathbb{R}^{3 \times 3 \times 1}$ ($C_{\text{out}} = 1$) with **Scalar Bias** $b = -1$ and **Activation Function** $\text{ReLU}(z) = \max(0, z)$:

$$\mathbf{X}_{:, :, 1} = \begin{bmatrix} 2 & 0 & 1 & 0 & 3 \\ 0 & 1 & 0 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 \\ 0 & 0 & 1 & 3 & 0 \\ 1 & 0 & 0 & 1 & 2 \end{bmatrix}, \quad \mathbf{K}_{:, :, 1} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Part A: Convolution Operation with Stride $s = 2$

Assuming stride $s = 2$ and padding $p = 0$, first compute the spatial and channel dimensions of the output feature map, then calculate all values of the output feature map $\mathbf{Y}$ after adding bias $b = -1$ and applying activation $\text{ReLU}(z) = \max(0, z)$.

- 1. **Output Dimensions ($H \times W \times C$):** _____
- 2. **Feature Map Calculations & Resulting Matrix:**

Part B: Max Pooling with Stride $s = 2$

Apply a 2×2 Max Pooling filter with stride $s = 2$ and padding $p = 0$ to the feature map $\mathbf{Y}$ you computed in Part A.

- 1. **Output Dimensions ($H \times W \times C$):** _____
- 2. **Feature Map Calculations & Resulting Matrix:**